

MT3054 Characterization of Materials

Semester 5

**A Case Study X-Ray Diffraction Analysis of Paper
Samples: Effects of Water Deacidification
Treatments and Artificial Ageing on Cellulose
Crystallinity**

By

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1. Introduction

The preservation of historical documents and archival paper materials requires careful consideration of their chemical and physical stability over time. Conservation treatments, particularly deacidification processes, are widely used to mitigate the effects of environmental deterioration. Previous studies have primarily focused on macroscopic changes in paper properties, such as pH, color, and mechanical strength. However, the effects of conservation treatments at the nano structural level, particularly on cellulose crystallinity, remain underexplored.

This study investigates the effects of water deacidification treatments and artificial ageing on the nano structural properties of cellulose in paper samples using X-ray diffraction (XRD). Crystallinity is a key factor influencing the long-term durability of cellulose-based materials, as it can affect their mechanical properties and susceptibility to further degradation. By employing XRD, this research aims to assess the extent to which crystallinity is influenced by conservation treatments and to provide insights into the preservation of paper over time. [1]

2. Sample Preparation

Two types of paper samples were selected for the study: old “rag” paper and new Whatman No. 1 paper. The old paper, dating back to 1831, primarily consists of flax fibers, while the Whatman No. 1 paper is composed of pure cotton fibers, serving as the modern reference material.

The preparation process for both samples included a washing step with deionized Millipore Elix III purified water, followed by deacidification treatments. Two different deacidification solutions were used: 0.02 M calcium hydroxide ($\text{Ca}(\text{OH})_2$) and 0.04 M magnesium bicarbonate ($\text{Mg}(\text{HCO}_3)_2$). The paper samples were submerged in the deacidification solutions at a ratio of 1 g of paper per 100 ml of solution, with each treatment lasting 30 minutes. After treatment, the samples were artificially aged in a climate chamber at 80°C and 65% relative humidity for two weeks to simulate the long-term ageing process that occurs naturally in historical documents.

These treated and artificially aged samples were then analyzed using X-ray diffraction (XRD) to assess the impact of these processes on the crystallinity of the cellulose fibers. [1]

3. Justification for the Use of X-ray Diffraction

X-ray diffraction (XRD) was chosen as the primary characterization method in this study due to its effectiveness in investigating the nano structural properties of cellulose in paper. XRD provides valuable insights into the crystallinity of cellulose, which is a critical factor in determining the paper's mechanical strength and its durability over time. By focusing on crystallinity, this study aims to assess how treatments such as washing and deacidification affect the structural integrity of paper.

The XRD technique enables the detection of both crystalline and amorphous regions within cellulose fibers, which is essential for understanding the paper's response to ageing and conservation treatments. The wide-angle X-ray diffraction (WAXD) system used in this research, located at Cardiff University, is equipped with a high-precision setup that allows the detection of features as small as 0.22 to 3.0 nanometers in cellulose structures. The X-ray beam, generated by a Kristalloflex 760 X-ray generator (Bruker AXS, Germany), produces detailed diffraction patterns that reveal the structural changes occurring in the paper samples.

Two key values from the diffraction data—the intensity of the crystalline cellulose peak (I_{max}) and the intensity of the amorphous scattering background (I_{min})—were used to calculate the Crystallinity Index (CI) and Crystallinity Ratio (CR). These metrics provide a quantitative measure of the paper's crystallinity, allowing for a detailed comparison between treated and untreated, aged, and unaged samples.

The use of XRD in this study is particularly relevant, as it allows for non-destructive analysis of fragile historical paper samples, preserving their integrity while yielding critical insights into their structural properties.

4. Results and Discussion

The results of the X-ray diffraction (XRD) analysis provided detailed insights into the crystallinity of both the old and new paper samples, with and without conservation treatments. Crystallinity was assessed using two key metrics: the Crystallinity Index (CI) and Crystallinity Ratio (CR). The calculated values for these metrics are presented in tables, highlighting the impact of washing, deacidification, and artificial ageing on the paper samples.

4.1 Old Paper Samples

For untreated old paper samples, the Crystallinity Index (CI) ranged between 0.660 and 0.668, while the Crystallinity Ratio (CR) ranged from 0.484 to 0.504. After washing with deionized water, the crystallinity slightly increased, with CI values of 0.669 to 0.672 and CR values of 0.506 to 0.512. Treatment with calcium hydroxide ($\text{Ca}(\text{OH})_2$) resulted in a modest increase in crystallinity, with CI values reaching 0.667 to 0.689 and CR values ranging from 0.502 to 0.549. However, treatment with magnesium bicarbonate ($\text{Mg}(\text{HCO}_3)_2$) produced mixed results, with a lower CI of 0.586 to 0.651 and a CR of 0.293 to 0.463, indicating that this treatment was less effective in promoting crystallinity.

Old Paper samples	CI	CR
Reference (untreated)	0.660	0.484
Reference (untreated)	0.668	0.504
Washed in deionised H_2O	0.669	0.506
Washed in deionised H_2O	0.672	0.512
Treated with $\text{Ca}(\text{OH})_2$	0.667	0.502
Treated with $\text{Ca}(\text{OH})_2$	0.689	0.549
Treated with $\text{Mg}(\text{HCO}_3)_2$	0.586	0.293
Treated with $\text{Mg}(\text{HCO}_3)_2$	0.651	0.463

Table 1 : Crystallinity Index and Crystallinity Ratio of old paper samples

The artificially aged old paper samples showed a significant variation in crystallinity depending on the treatment. Untreated aged samples had a CI between 0.714 and 0.721, while the CR ranged from 0.600 to 0.613. Washing with deionized water resulted in a noticeable decrease in crystallinity, with CI values around 0.671 to 0.676 and CR values between 0.510 and 0.522. Calcium hydroxide treatment maintained similar crystallinity levels, with CI values of 0.670 to 0.671 and CR values around 0.507 to 0.510. Magnesium bicarbonate treatment led to a higher crystallinity in aged samples, with a CI of 0.670 to 0.704 and a CR of 0.507 to 0.580.

Aged Old Paper samples	CI	CR
Reference (aged)	0.721	0.613
Reference (aged)	0.714	0.600
Washed in deionised H ₂ O	0.671	0.510
Washed in deionised H ₂ O	0.676	0.522
Treated with Ca(OH) ₂	0.671	0.510
Treated with Ca(OH) ₂	0.670	0.507
Treated with Mg(HCO ₃) ₂	0.704	0.580
Treated with Mg(HCO ₃) ₂	0.670	0.507

Table 2 : Crystallinity Index and Crystallinity Ratio of aged old paper samples

A key finding from the XRD analysis of old paper samples treated with calcium hydroxide was the presence of calcite deposits. This was observed as a distinct reflection in the diffraction patterns, indicating the formation of calcite after treatment. Figure 2 shows the X-ray diffraction image of the old paper treated with calcium hydroxide, where the calcite reflection has been labeled.

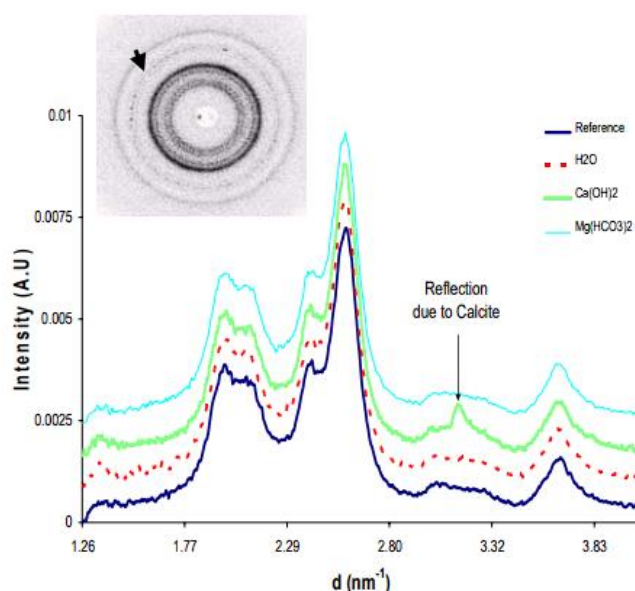


Figure 2 : X-ray diffraction image of old paper treated with Ca(OH)₂ showing calcite reflection

4.2 New Paper Samples

The new Whatman paper samples displayed higher overall crystallinity compared to the old rag paper samples. Untreated new paper had CI values of 0.721 to 0.725 and CR values between 0.612 and 0.621. Calcium hydroxide treatment slightly increased the crystallinity, with CI values of 0.729 to 0.732 and CR values between 0.629 and 0.634. Treatment with magnesium bicarbonate showed minimal change, with CI values ranging from 0.720 to 0.723 and CR values from 0.611 to 0.618.

New Paper samples	CI	CR
Reference (untreated)	0.725	0.621
Reference (untreated)	0.721	0.612
Treated with $\text{Ca}(\text{OH})_2$	0.732	0.634
Treated with $\text{Ca}(\text{OH})_2$	0.729	0.629
Treated $\text{Mg}(\text{HCO}_3)_2$	0.723	0.618
Treated $\text{Mg}(\text{HCO}_3)_2$	0.720	0.611

Table 3 : Crystallinity Index and Crystallinity Ratio of new paper samples

After artificial ageing, the crystallinity of new paper samples increased further. Aged untreated samples had CI values of 0.742 to 0.744 and CR values of 0.652 to 0.655. Treatment with calcium hydroxide slightly enhanced crystallinity in aged samples, with CI values reaching 0.732 to 0.751 and CR values of 0.635 to 0.668. Magnesium bicarbonate treatment resulted in stable crystallinity levels in aged samples, with CI values between 0.735 to 0.742 and CR values between 0.639 to 0.652.

Aged New Paper samples	CI	CR
Reference (aged)	0.744	0.655
Reference (aged)	0.742	0.652
Treated with $\text{Ca}(\text{OH})_2$	0.732	0.635
Treated with $\text{Ca}(\text{OH})_2$	0.751	0.668
Treated with $\text{Mg}(\text{HCO}_3)_2$	0.742	0.652
Treated with $\text{Mg}(\text{HCO}_3)_2$	0.735	0.639

Table 4 : Crystallinity Index and Crystallinity Ratio of aged new paper samples

4.3 Discussion

The results indicate that the ageing process generally increases the crystallinity of both old and new paper samples. Untreated new paper showed higher crystallinity compared to old paper, likely due to differences in fiber composition (cotton vs. flax). The deacidification treatments, particularly calcium hydroxide, appeared to have a protective effect on crystallinity in old paper samples, preventing significant crystallization during ageing. In contrast, magnesium bicarbonate was less effective in maintaining crystallinity levels in both new and old paper samples.

A key finding from the XRD analysis was the presence of calcite deposits in the old paper samples treated with calcium hydroxide. This was evidenced by a distinct diffraction pattern, suggesting that calcium hydroxide leaves residual calcite, which may influence the overall structural properties of the treated paper. This observation highlights the importance of considering the potential side effects of conservation treatments on historical materials.

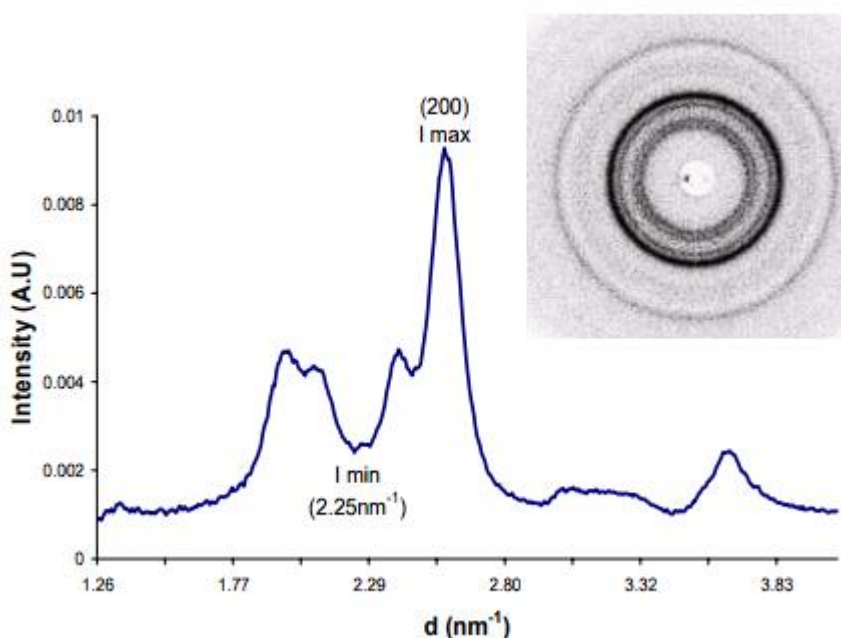


Figure 2 : X-ray diffraction image of Whatman paper sample

5. Conclusion

This study utilized X-ray diffraction (XRD) to investigate the effects of aqueous conservation treatments and artificial ageing on the crystallinity of cellulose in both old rag and new Whatman paper samples. The results revealed that the ageing process generally increased the crystallinity of both paper types, with new paper displaying higher crystallinity than old paper. This difference is attributed to the variations in fiber composition, with new paper consisting of cotton fibers and old paper composed of flax fibers.

The deacidification treatments, especially with calcium hydroxide ($\text{Ca}(\text{OH})_2$), were found to be effective in preventing further crystallization in aged samples of old rag paper, thereby offering a protective effect during the ageing process. However, the magnesium bicarbonate ($\text{Mg}(\text{HCO}_3)_2$) treatment yielded less consistent results, particularly in aged old paper samples, where it was less effective in maintaining crystallinity. Additionally, calcium hydroxide treatment left residual calcite deposits, as evidenced by the XRD diffraction patterns, which may impact the structural integrity of the treated samples.

These findings underscore the importance of selecting appropriate conservation treatments for historical paper materials to preserve their nano structural integrity and prevent excessive crystallization, which could compromise their mechanical properties over time. XRD has proven to be a valuable tool for assessing the nano structural effects of these treatments, offering insights into how conservation methods can influence the long-term durability of cellulose-based materials.

6. References

[1] U. Knuutinen, C. J. Kennedy, T. J. Wess, and A. C. Maxwell, "X-Ray diffraction analysis of paper samples – Investigation into the effects of water deacidification treatments and artificial ageing process on cellulose crystallinity," *ResearchGate*, 2008. [Online]. Available: [https://www.researchgate.net/profile/Ulla-Knuutinen/publication/259927491_X-Ray_diffraction_analysis_of_paper_samples -
Investigation into the effects of water deacidification treatments and artificial ageing process on cellulose crystallinity/links/5581850c08ae12bde6e4a615/X-Ray-diffraction-analysis-of-paper-samples-Investigation-into-the-effects-of-water-deacidification-treatments-and-artificial-ageing-process-on-cellulose-crystallinity.pdf](https://www.researchgate.net/profile/Ulla-Knuutinen/publication/259927491_X-Ray_diffraction_analysis_of_paper_samples_-_Investigation_into_the_effects_of_water_deacidification_treatments_and_artificial_ageing_process_on_cellulose_crystallinity/links/5581850c08ae12bde6e4a615/X-Ray-diffraction-analysis-of-paper-samples-Investigation-into-the-effects-of-water-deacidification-treatments-and-artificial-ageing-process-on-cellulose-crystallinity.pdf). [Accessed: Sep. 9, 2024].